KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 7 REPORT

Dimensionality Reduction

Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

Team Member Name	Roll Number
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1. Week 7 Contribution

Implemented correlation analysis, feature selection and Principal Component Analysis (PCA) as part of the dimensionality-reduction stage for the Youth Smoking and Drug Consumption Analysis project.

2. Work Carried Out

The Week 7 work focused on the seven personality features in the cleaned dataset: Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS. A separate feature dataset was created from the cleaned data. The notebook reports 1,885 records and 7 selected features.

3. Dimensionality Reduction Activities

3.1 Feature Selection and Variance Thresholding

The seven personality variables were selected for analysis. Variance Thresholding was applied with a threshold of 0.01 to identify features with very low variance. All seven features had variance values above the threshold, so no feature was removed.

Feature	Variance
Nscore	0.942604
Escore	0.994903
Oscore	0.992471
Ascore	0.994888
Cscore	0.995051
Impulsive	0.910945
SS	0.928720

3.2 Correlation-Based Feature Selection

A Pearson correlation matrix was calculated for the seven personality features and visualized using a heatmap. The notebook used an absolute correlation threshold greater than 0.80 to flag highly correlated pairs. No feature pair crossed this threshold; therefore, no pair was removed through this correlation-based check.

The strongest positive relationship shown was between Impulsive and SS (0.623120). The strongest negative relationship shown was between Nscore and Escore (-0.414434). These values describe association, not causation.

3.3 Multicollinearity Analysis using VIF

The features were standardized before calculating the Variance Inflation Factor (VIF). All reported VIF values were below 5, and the notebook produced an empty list for features requiring further investigation under the selected warning threshold.

Feature	VIF
Nscore	1.354701
Escore	1.471786
Oscore	1.280984
Ascore	1.150772
Cscore	1.389940
Impulsive	1.790268
SS	1.914818

3.4 Principal Component Analysis (PCA)

The seven original personality features were standardized using StandardScaler before PCA was applied. The initial PCA generated seven principal components. The explained-variance ratio was then examined to decide how many components were needed to retain at least 95% of the total variance.

3.5 Explained Variance Analysis

Principal Component	Explained Variance	Cumulative Variance
PC1	31.99%	31.99%
PC2	25.77%	57.76%
PC3	12.69%	70.45%
PC4	9.65%	80.11%
PC5	8.16%	88.26%
PC6	6.80%	95.07%
PC7	4.93%	100.00%

The scree and cumulative-variance visualization showed how the retained information increases as components are added. Six components were sufficient to cross the 95% target; the first six retained 95.07% of the variance.

3.6 Reduced Dataset and PCA Visualization

PCA was applied again with six selected components. The resulting reduced dataset contained 1,885 rows and 6 columns (PC1 to PC6), compared with the original 1,885 rows and 7 personality features. A two-dimensional scatter plot of PC1 against PC2 was also created, with points distinguished by the gender labels available in the cleaned dataset.

4. Outcome of My Contribution

The Week 7 analysis examined feature variance, pairwise correlations and multicollinearity before applying PCA. Variance Thresholding retained all seven features, no pair exceeded the correlation threshold of 0.80, and all VIF values were below 5. PCA reduced the seven-dimensional personality feature set to six principal components while retaining 95.07% of the variance. The reduced data shape was (1885, 6).